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Article

Targeted passivation and optimized interfacial carrier dynamics improving the efficiency and stability of hole transport layer-free narrow-bandgap perovskite solar cells

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ABSTRACT

Narrow-bandgap mixed Sn-Pb perovskite solar cells (PSCs) have showcased great potential to approach the Shockley-Queisser limit. Nevertheless, the practical application and long-term deployment of mixed Sn-Pb PSCs are still largely impeded by the rapid oxidation of Sn^{2+} ions and under-optimized carrier transport layer (CTL)/perovskite interfaces that would inevitably incur serious interfacial charge recombination and device performance degradation. Herein, we successfully removed the hole transport layer (HTL) by incorporating a small amount of organic phosphonic acid molecules into perovskites, which could preferably interact with Sn^{2+} ions (relative to Pb^{2+} analogues) at the grain boundaries (GBs) throughout the perovskite film thickness via coordination bonding, thus effectively retarding the oxidation of Sn^{2+}, passivating the defects and suppressing the non-radiative recombination. Targeted modification effectively reinforced built-in potential by ~100 mV, and favorably induced energy level cascade, thus accelerating spatial charge separation and facilitating the hole extraction from perovskite layer to underlying conductive electrodes even in the absence of HTL. Consequently, enhanced power conversion efficiencies up to 20.21% have been achieved, which is the record efficiency for the HTL-free mixed Sn-Pb PSCs, accompanied by a decent photovoltage of 0.87 V and improved long-term stability over 2400 h.

1. Introduction

Over the past decades, the single-junction organic–inorganic metal halide perovskite solar cells (PSCs) have witnessed enormous success and altered the pathway towards next-generation photovoltaic technologies. Up till now, the state-of-the-art best-performing PSCs have reached certified power conversion efficiencies (PCEs) up to 25.8% [1], which were mainly based on the utilization of Pb-based perovskites with bandgaps larger than 1.45 eV [2]. The emergence of narrow-bandgap mixed Sn-Pb perovskites featuring small bandgaps, i.e., ~1.2–1.3 eV, showcased vast potential to bring the PCEs of resultant PSCs even closer to the theoretical Shockley-Queisser (SQ) limit, and to some extent release the toxicity issue of perovskite components [3–5]. However, the mixed Sn-Pb PSCs normally suffer from low stability arising from the rapid oxidation of stannous ions, i.e., from Sn^{2+} to Sn^{4+}, which would result in high density of crystallographic defects, leading to unwanted non-radiative recombination and thus diminishing the photovoltaic performance of device [6–8]. Several antioxidant additives, such as Sn powder, hydroxybenzene sulfonic acid, SnF_2 and SnF_2-pyrazine complex, have been employed to simultaneously suppress the Sn^{2+} oxidation and passivate the defects in mixed Sn-Pb perovskite films [9–11]. Encouragingly, the best PCEs of the state-of-the-art mixed Sn-Pb PSCs have reached 23.7%, progressively narrowing the gap with their pure-Pb counterparts [12,13].

Another common issue that affects the practical application and long-term deployment of mixed Sn-Pb PSCs is the limited options

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of suitable hole transport materials (HTMs), e.g., inorganic nickel oxide (NiO_x), poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA) and poly (3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS). These HTMs were either extremely expensive or not perfectly compatible with mixed Sn-Pb perovskites. Worse still, in some cases, they would trigger the degradation or accelerate the decomposition of perovskite layers, as well as cause severe trap-assisted recombination at the hole transport layer (HTL)/perovskite interface.

Rather than exploring more suitable and economic HTMs, developing HTL-free mixed Sn-Pb PSCs provides an appealing option for fabricating high-efficiency and high-stability Sn-based devices in a low-cost manner. Ning and co-workers [14] reported the HTL-free mixed Sn-Pb PSCs in 2017. By systematically optimizing the Pb-to-Sn ratio and the concentration of SnF_2, a PCE of 7.94% was obtained for $FAPb_{0.5}Sn_{0.5}I_3$-based HTL-free PSCs. However, studies on HTL-free mixed Sn-Pb PSCs are still scarce. Indeed, the Sn-containing perovskites intrinsically feature p-type doping characteristic, which is advantageous to produce HTL-free PSCs, but only with moderate efficiencies [15,16]. Up till now, the reported strategies to fabricate HTL-free mixed Sn-Pb PSCs were still limited, which mainly focused on improving the antioxidant capabilities of mixed Sn-Pb perovskites by using formamidinesulfinic acid (FSA) or binary additive system [17,18]. But the photovoltaic performance of HTL-free devices still far lags behind the conventional counterparts with HTL. Learned from the successful cases of pure Pb-based devices without charge transport layer (CTL) [19,20], the key to constructing efficient HTL-free PSCs with inverted p-i-n structure is to alleviate the interfacial energy-level mismatch between perovskite layer and underlying conductive electrodes, as a large energy barrier is considered to largely inhibit the interfacial carrier extraction and collection [21–23]. Unfortunately, this aspect has been neglected in the realm of HTL-free mixed Sn-Pb PSCs.

In this work, we report a novel and peculiar "all-in-one" targeted passivation and carrier dynamic modulation strategy to fabricate the HTL-free mixed Sn-Pb PSCs. By deliberately selecting an organic molecule, [2-(9H-carbazol-9-yl) ethyl] phosphonic acid (2PACz), comprising conjugated moieties and Lewis base-typed functional groups [24], we are able to selectively anchor the molecule at the grain boundaries (GBs) of perovskite film and targetedly interact with undercoordinated Sn^{2+} in mixed Sn-Pb perovskites, thus effectively inhibiting Sn^{2+} oxidization and passivating the crystallographic defects. In addition, the 2PACz modification could simultaneously reinforce the built-in potential by ~100 mV for boosted charge separation within the bulk perovskite, and more importantly, induce a favorable energy level cascade for boosting carrier extraction and collection from the perovskite layer to indium-doped tin oxide (ITO) electrode even in the absence of HTL. As a result, a champion PCE of 20.21% was achieved for the HTL-free PSCs based on $Cs_{0.05}MA_{0.45}FA_{0.5}Pb_{0.5}Sn_{0.5}I_3$ perovskite film with 2PACz-assisted targeted modification, accompanied by greatly improved device stability over 2400 h. This work endows an effective wet-chemical approach to fabricate the mixed Sn-Pb PSCs with simplified device architecture, facilitating the commercial application of environmentally viable Sn-based photovoltaics in the near future.

2. Experimental

2.1. Materials

Formamidinium iodide (99.99%) (FAI) and methylammonium iodide (99.99%) (MAI) were purchased from Greatcell Solar company. *N*,*N*-dimethylmethanamide (99.8%) (DMF), dimethylsulfoxide (99.9%) (DMSO) and chlorobenzene (99.8%) were purchased from Advanced Election Technology. SnI_2 (99.99%), SnF_2 (99%) and bathocuproine (BCP) were obtained from Sigma-Aldrich. CsI (99.999%) and C_{60} were purchased from Xi'an Polymer Light Technology. PbI_2 (99.999%) was obtained from Alfa Aesar. 2PACz was purchased from TCI (Shanghai) Development. All chemicals were used as received without further purification.

2.2. Perovskite ink preparation

For the $Cs_{0.05}MA_{0.45}FA_{0.5}Pb_{0.5}Sn_{0.5}I_3$ perovskite, the precursor solution (1.6 mol/L) was prepared in mixed solvents of DMF and DMSO with a volume ratio of 7:3. SnF_2 (10 mol%, relative to SnI_2) was added to the perovskite precursor solution. 2PACZ solution (1 mol/L in DMF) was prepared as additive for perovskite precursors, and the 2PACz concentrations have been varied from 0.05 mol% to 0.20 mol% in perovskite. The perovskite solution preparation was conducted in a nitrogen glove box with 0.1 ppm oxygen and <1 ppm H_2O.

2.3. Device fabrication

Patterned ITO glass substrates were cleaned by sequential ultrasonication in deionized (DI) water, acetone and isopropyl alcohol for 25 min, respectively. After drying, the ITO substrates were further treated via UV-ozone treatment for 25 min. The mixed Sn-Pb perovskite precursor was spin-coated on the ITO substrate for 50 s and 200 μL of chlorobenzene was dropped on the film after 20 s elapsed. The as-prepared perovskite films were further treated by ethyl acetate solvent washing to remove excessive 2PACz assembled on the top surface, which to a large extent got rid of the direct contact of the 2PACz with the electron transport layer (ETL). The substrates were then transferred onto a hotplate and annealed at 100 °C for 10 min. To fabricate the PSCs based on the independent 2PACz HTL, 2PACz (0.2–0.5 mg/mL in isopropanol) was deposited onto ITO substrates via spin-coating at 3000 r/min for 30 s, followed by heating at 100 °C for 10 min. Finally, 20 nm C_{60}, 7 nm BCP and 80 nm Cu were sequentially thermal evaporated on top of the perovskite films to complete the device.

2.4. Material and device characterization

The X-ray diffraction (XRD) pattern was obtained by using a Miniflex600 X-ray diffractometer. The UV–vis absorption (equipped with an integrating sphere to exclude signal due to light scattering) was obtained from the UV-3600 spectrophotometer (Shimadzu). The morphological features of the samples were characterized using a field emission scanning electron microscope (SEM, Hitachi-SU8010). Atomic force microscopy-based infrared (AFM-IR) spectroscopy data were collected on the nanoIR2-s-AFMIR. Kelvin Probe Force Microscopy (KPFM) was carried on a Scanning Probe Microscope (Bruker Dimension Fastscan). For the KPFM characterization, the perovskite films were exfoliated from the substrates and flipped to uncover the bottom surface, with the other side attached to conductive copper tape. The *J*–*V* characteristics of the PSCs were measured using a Keithley 2400 Sourcemeter under the simulated AM1.5G one sun illumination (100 mW/cm^2) generated by a solar simulator (the Oriel Sol3A Class AAA) based on Xenon lamps. The *J*–*V* curves were measured under reverse or forward scanning directions in the range of −0.1–0.9 V at a scan rate of 100 mV/s (a voltage step of 10 mV and a delay time of 10 ms) in ambient air ((25 ± 5) °C and 35% ± 10% relative humidity). The light intensity was calibrated by a silicon reference cell equipped with a SCHOTT visible color KG5 glass filter (Newport 91150 V). Non-reflective shadow masks with an aperture of 0.08 cm^2 were used to define the active regions of the PSCs. For the thermal stability test, the devices without encapsulation are heated at 60 °C in N_2-filled glovebox for different durations. For the light-stability test, the devices with encapsulation were connected to a resistor to make them operate at the maximum power point (MPP) and soaked under continuous one-sun illumination. X-ray photoelectron spectra (XPS) were investigated on an ESCALAB 250 photoelectron spectrometer (Thermo Fisher Scientific). Nuclear magnetism resonance (NMR) spectra were measured on Bruker Advance spectrometers (500 MHz for ^{1}H NMR and 202.4 MHz for ^{31}P NMR). To investigate the defect profile in the perovskite films, the space-charge limited current (SCLC) test was performed using a CH Instruments CHI660 electrochemical workstation in a shield box with the configuration of ITO/PEDOT:PSS/perovskite/Spiro-OMeTAD/Cu. An Edinburgh Instruments Ltd FLSP1000 was used to test the steady-state photoluminescence (PL) and time-resolved photoluminescence (TRPL) with a pulsed excitation laser of 460 nm.

2.5. Femtosecond transient absorption (fs-TA) spectroscopy characterization

The fs-TA measurement was performed by equipping a regeneratively amplified Ti: Sapphire laser source (Coherent Legend, 800 nm, 150 fs, 5 mJ/pulse, and 1 kHz repetition rate) and Helios (Ultrafast Systems LLC) spectrometers. A 75% portion of the 800 nm output pulse was frequency-doubled in the barium borate (BaB_2O_4) crystal, which could produce 400 nm pump light. The rest portion of the output was concentrated using a sapphire window to generate a near-infrared light continuum (820–1600 nm) probe light. The 400 nm pump beam was generated as part of the output pulse from the 800 nm amplifier, and its power was adjusted by a set of neutral-density filters. By applying complementary metal oxide–semiconductor sensors with a frequency of 1 kHz, the probe and reference beams could be separated from the white-light continuum and sent into a fiber optics-coupled multichannel spectrometer.

2.6. Density functional theory calculation

Density functional theory calculations were conducted in Vienna *ab-initio* Software Package in version 5.4.1 [25,26]. The electronic exchange–correlation was described by the Perdew-Burke-Ernzerhof formula under the generalized gradient approximation with the van der Waals correction [27,28]. The kinetic cut-off energy was set to 520 eV and the threshold of energy and force during geometry optimizations were corresponding to 10^{-6} eV and -0.01 eV/Å, respectively.

3. Results and discussion

Mixed Sn-Pb perovskite, i.e., $Cs_{0.05}MA_{0.45}FA_{0.5}Pb_{0.5}Sn_{0.5}I_3$ with a bandgap of 1.22 eV [29], was selected as the light absorber for detailed investigation. Incorporation of functional molecules into perovskite inks and manipulation of relevant chemical interaction have been regarded as effective strategies to modulate the optoelectronic properties (i.e., doping types, trap densities, energy levels, etc.) and anti-oxidation capabilities of mixed Sn-Pb perovskites [15]. For the Sn-Pb alloyed perovskites, the Sn^{2+} ions play a critical role in determining the bandgap, defect density, doping level, crystallization kinetics and stability of resultant perovskite films, and hence the targeted passivation and stabilization of Sn^{2+} by rationally designed heterogeneous molecules is of great significance [30–32]. Herein, a considerable amount of 2PACz has been directly added into perovskite ink. Typically, the multidentate 2PACz molecule (Fig. 1a) is composed of diverse functional groups, namely, the carbazole group that could facilitate electron delocalization and the phosphonic acid (PA) tail that could share the lone-pair electrons to form chemical complex with metal halides moieties in perovskites [33–35].

Fig. 1a illustrated the electrostatic potential (ESP) of different functional groups in 2PACz. Compared with the P-OH counterpart, the P═O bond exhibited a more negative ESP, corresponding to a higher electron density and thus a stronger electron donation capability. In general, the electron-donating group (P═O) could both interact with undercoordinated Pb^{2+} or Sn^{2+}. Further calculation and analysis indicated a stronger coordination ability of 2PACz with Sn^{2+}, relative to Pb^{2+} (i.e., a binding energy of -33.67 kcal/mol for 2PACz-SnI_2 vs. -30.90 kcal/mol for 2PACz-PbI_2, Fig. 1b), suggesting the unique and desirable targeted interaction and passivation functions of 2PACz towards Sn^{2+} species, which could be ascribed to the relatively large electronegativity of Sn atom over Pb analogue. A schematic illustration of chemical interaction between 2PACz and mixed Sn-Pb perovskite was depicted in Fig. 1c. Specifically, the 2PACz could favorably interact and passivate the under-coordinated metal ions, especially the Sn^{2+} ions in perovskites, in a way of forming coordination bonds by sharing the lone electron pair at O atom of the phosphonic group in 2PACz with the empty 5p orbit of Sn^{2+} [36].

To authenticate the potential chemical interaction between metal ions and 2PACz, Fourier transform infrared spectroscopy (FTIR) was conducted for both PbI_2 and SnI_2 films modified with 2PACz, as well as for the pure 2PACz film. The pure 2PACz sample exhibited a characteristic vibration band of P═O bond at 1178 cm^{-1}. For the SnI_2-2PACz sample, a more obvious shift of P═O vibration band (1165 cm^{-1}) was observed, relative to PbI_2-2PACz counterpart (1169 cm^{-1}), suggesting a more robust chemical interaction between 2PACz and SnI_2 (Fig. 1d). Moreover, liquid-state nuclear magnetic resonance (NMR) spectroscopy measurements of 2PACz, 2PACz-PbI_2 and 2PACz-SnI_2 in DMSO d_6 solution were conducted to identify the coordination interaction. In accordance with the phenomenon observed in FTIR, the ^{31}P resonances (Fig. S1 online) exhibited clear differences between 2PACz-PbI_2 and 2PACz-SnI_2 solutions. Compared to the pure 2PACz, the ^{31}P resonances of 2PACz-PbI_2 barely exhibited any variation, suggesting that the interaction between 2PACz and PbI_2 might be below the detection limit. By contrast, the ^{31}P resonances of 2PACz-SnI_2 complex were distinctly different from that of neat 2PACz. In addition to the peak originating from the native 2PACz, there appeared a new set of peaks possibly related to the PA group and other complex derivatives in a new environment, which could be attributed to the direct and preferable interaction between SnI_2 and PA group. Similarly, the ^{1}H spectrum of 2PACz-SnI_2 complex which represented the molecular fragment of carbazole was split and broadened (Fig. S1 online). The new hydrogen environment could arise from the interaction between the 2PACz and SnI_2, which, instead, was not observed in the 2PACz-PbI_2 complex. Accordingly, we conclude that there exists a comparatively stronger force between 2PACz and SnI_2 than that between 2PACz and PbI_2, again verifying the targeted coordination function of the 2PACz additive on Sn-based perovskites.

To further understand the interaction between perovskite and 2PACz additive, XPS meansurement was performed. Upon 2PACz modification, the binding energy of Pb $4f_{7/2}$ peak and Sn $3d_{5/2}$ peak both shifted to higher positions, but to a different extent. Specifically, a red-shift of 0.24 eV (from 486.70 to 486.94 eV, Fig. 1f) has been observed for the Sn $3d_{5/2}$ peak, which is greater than that of Pb $4f_{7/2}$ peak (0.16 eV, from 138.03 to 138.19 eV, Fig. 1e), again confirming stronger bonding or coordination of 2PACz with Sn^{2+}, thus significantly affecting the electron cloud density.

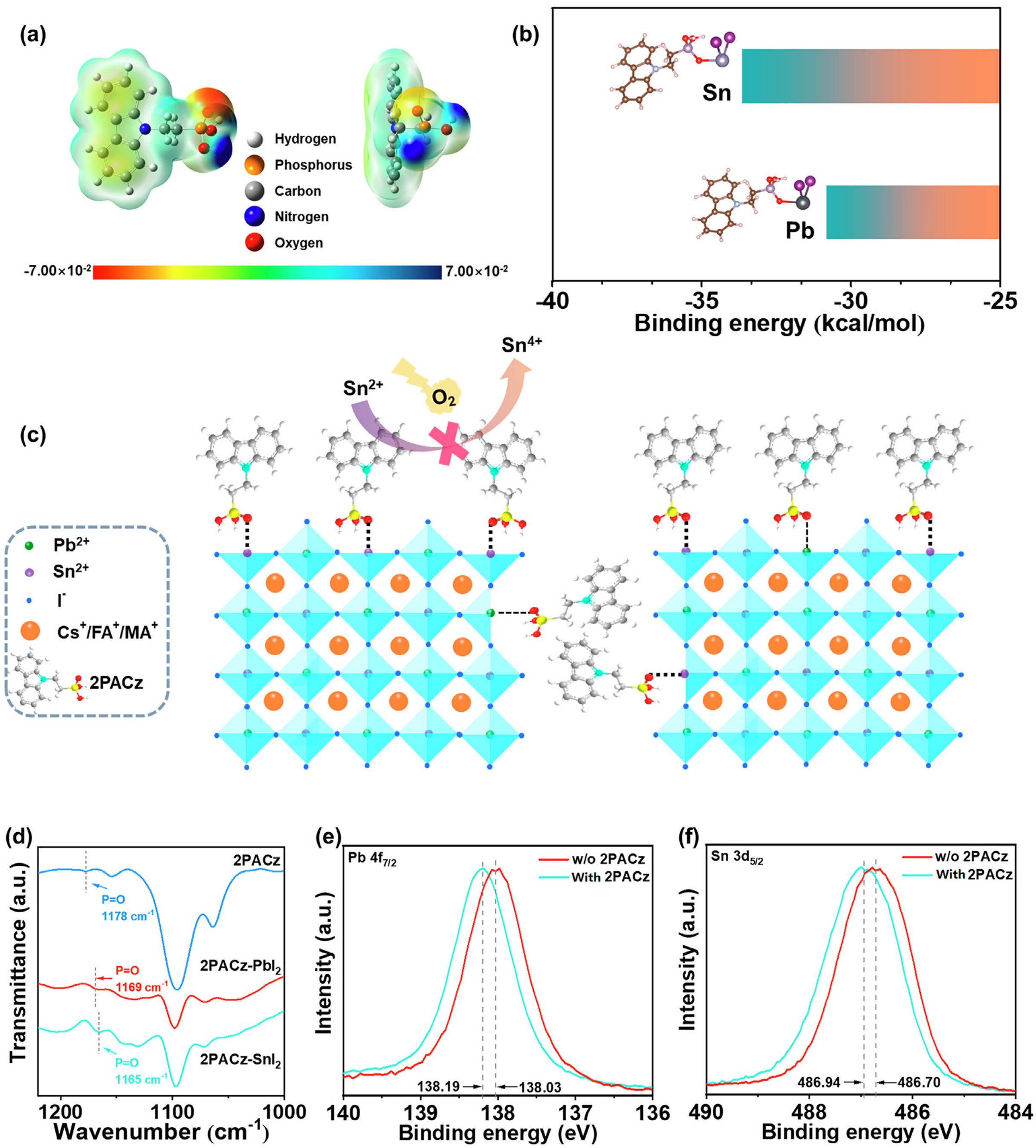

Fig. 1. (Color online) Interaction between 2PACz and SnI_2/PbI_2. (a) Molecular structure and calculated ESP profile of 2PACz molecule. (b) Interaction energies of 2PACz with SnI_2 and PbI_2. (c) Schematic illustrations of targeted interaction, passivation and stabilization of mixed Sn-Pb perovskites by 2PACz molecules. (d) FTIR spectra of 2PACz, 2PACz-PbI_2 complex and 2PACz-SnI_2 complex. XPS spectra of (e) Pb $4f_{7/2}$ and (f) Sn $3d_{5/2}$ for the perovskite films with or without 2PACz modification.

To appraise the antioxidation effect of the 2PACz additive on Sn^{2+}, the perovskite inks with or without 2PACz were prepared, followed by exposure to air for 20 min. At the very beginning, both of the solutions exhibited a bright-yellow color. The perovskite ink without 2PACz completely turned dark-red color in 20 min, indicating a large portion of Sn^{2+} was rapidly oxidized to Sn^{4+}. By contrast, the color of perovskite precursor with 2PACz addition only showed some orange color on the very top surface for the same duration, implying the positive role of 2PACz in retarding the oxidation of Sn^{2+} (Fig. S2 online). XPS study was also used to analyze the antioxidation ability of 2PACz additive. Typically, there was P 2p signal showing up at 133.3 eV, verifying the presence of 2PACz additive in the modified perovskite films (Fig. S3 online). We also conducted the XPS characterization on the bottom surface of perovskite film exfoliated from the substrate (Fig. S4 online), which also occasionally showed the presence of P 2p signals (Fig. S5 online), suggesting the spontaneous segregation of 2PACz molecules onto the ITO/perovskite buried interface [37]. However, we deduced that the 2PACz was just partially distributed at the buried interface, rather than self-assembling to a uniform monolayer, considering the XPS is a surface-sensitive technique and the P 2p signals were not always detectable in every 2PACz-modified sample or at different spots of the same 2PACz-modified sample. Specifically, the Sn 3d signal can be fitted and divided into Sn^{2+} peak and Sn^{4+} peak, respectively. When 2PACz was introduced, the Sn^{4+}/Sn^{2+} ratio decreased significantly in mixed Sn-Pb perovskite films (Fig. 2b), again verifying the favorable capability of 2PACz on protection and stabilization of Sn^{2+} (Fig. 1c). By contrast, the pristine perovskite film without 2PACz incorporation exhibited low stability and showcased fast oxidization of Sn^{2+} under identical conditions (Fig. 2a).

The crystallization of Sn-containing perovskites is notorious to be too fast and thus hard to be well-controlled, which easily leads to the formation of low-quality perovskite film with inferior coverage. In this regard, additive engineering has been regarded as a useful technique to modulate the crystallization kinetics of perovskites, which is conducive to achieving a uniform perovskite film with improved quality [38,39]. Herein, we speculated that the asymmetrical coordination effect of 2PACz on Sn^{2+} and Pb^{2+}, for instance, more preferable interaction with Sn^{2+}, would postpone the crystallization via retarding the fast reaction of SnI_2 and organic ammonium halide, as well as facilitate homogeneous nucleation and crystal growth of the mixed Sn-Pb perovskites, which is advantageous to acquire highly crystalline and uniform perovskite film. To determine the effect of 2PACz additive on the crystal structure and crystallinity of mixed Sn-Pb perovskite films, we conducted XRD measurements for the control and 2PACz-modified samples, respectively. As shown in Fig. 2c, a series of diffraction peaks attributed to the (1 1 0), (2 0 2), (2 2 0) and (3 1 0) crystal planes of the pseudocubic crystal structure were detected for both perovskite films. Notably, introducing 0.1 mol% 2PACz did not result in any observable XRD peak shift, suggesting the 2PACz with large molecular size could not be fitted into the perovskite crystal lattice [40–42]. One can also notice 2PACz incorporation had negligible effects on the absorption property and optical bandgap of the resultant perovskite films (Fig. S6 online). In addition, all the characteristic XRD peaks for the 2PACz-modified perovskite film were reinforced, relative to the control perovskite film, suggesting improved crystallinity of perovskite film with 2PACz modification. The effect of 2PACz additive on the morphologies of the perovskite films was also explored. Obviously, there were sheet-like species (white precipitations) distributed at the GBs for the control sample without 2PACz, which could be ascribed to PbI_2/SnI_2 impurities, as is also evidenced by the occurrence of additional XRD peak around 12.6°. Excess metal halides have been proven to be detrimental to the stability of PSC devices, especially under continuous illumination stress. However, when a small amount (i.e., 0.1 mol%) of 2PACz was introduced, these impurities disappeared, which was in accordance with the XRD results (Fig. S7 online). Generally, during the fabrication of perovskite films, PbI_2/SnI_2 would coordinate with DMSO and form complexes via dative bonding. When the perovskite film was annealed, DMSO quickly evaporated and PbI_2/SnI_2 would rapidly bind with other perovskite constituents, thus accelerating the crystallization of perovskite film, normally accompanied by a high density of defects [7]. In our work, 2PACz with electron-donating groups could intensively interact with divalent metal ions via strong coordination bonding, which would promote homogeneous nucleation, slow down the crystallization rate of mixed Sn-Pb perovskites, facilitate the reaction of raw materials and maintain the phase purity of resultant perovskite films.

To further trace the spatial distribution of 2PACz in the mixed Sn-Pb perovskite films, AFM-IR spectroscopy was performed, which is highly sensitive to the chemical identification of samples at a scale of tens of nanometers [43]. The AFM topography images and corresponding IR absorption images of both control and 2PACz-modified perovskite films were illustrated in Fig. S8 (online) and Fig. 2d,e, respectively. The IR absorption image was collected at 1032 cm^{-1} corresponding to P-containing bond of 2PACz. The bright red and yellow-color region only existed in the targeted perovskite film, which reflected the high IR vibrational activity of 2PACz. Overlapping with the AFM topography image, the spatial distribution of the yellow/red signals apparently appeared at GBs between adjacent grains (Fig. 2d and e), proving the 2PACz with large molecular size primarily condensed at the GBs or grain surface of the modified perovskite films. To further confirm the spatial distribution of 2PACz molecules across the entire perovskite film, the time-of-flight secondary ion mass spectrometry (TOF-SIMS) measurement was performed. For the target perovskite film, a pronounced phosphorous (P) signal from 2PACz molecule can be observed throughout the film, which was drastically intensified near the ITO surface, suggesting the tendency of 2PACz to aggregate at the buried interface between ITO substrate and perovskite film (Fig. S9 online). We speculated that such peculiar distribution of 2PACz could be attributed to its large molecular size and stronger coordination ability with Sn ions, which is also one of the fundamental components in ITO electrodes. We concluded that the 2PACz could self-anchor on the crystal periphery of mixed Sn-Pb perovskites and purposely interact with under-coordinated Sn^{2+} via robust coordination interaction and/or electrostatic attraction, which is conducive to passivating surface defects. In addition, such a robust chemical interaction could stabilize and effectively protect Sn^{2+} against oxygen invasion (Fig. 1c), which is helpful in reducing the formation of Sn vacancies.

Apart from the abovementioned targeted passivation and stabilization function, the 2PACz itself could act as a highly efficient hole selective contact owing to its suitable energy level alignment with various perovskite compositions [12,44–46]. In this regard, the 2PACz could serve as a suitable dopant in perovskites, which is capable of altering the work function of mixed Sn-Pb perovskites in a desirable way, thus making it possible and feasible to simplify the device structure [34,35,44]. We visualized the 2PACz molecules just partially covered on the grain surface and mainly condensed at GBs (Fig. 2e). To elucidate the energy level modulation effect of 2PACz on the perovskites, ultraviolet photoelectron spectroscopy (UPS) was characterized for both the control and 2PACz-modified perovskite films (Fig. S10 online). Note, in order to better reflect the effect of 2PACz modification on the interfacial properties of ITO/perovskite interface in the absence of HTL, the UPS measurement was conducted on the bottom surface of perovskite films which were peeled off from the substrate (Fig. S4 online) [20]. As a result, the Fermi level (E_F) negatively shifted from −4.89 eV for the control sample to −5.01 eV for the 2PACz-modified one. Moreover, the depth-profiling UPS test was conducted for the control and target perovskite films in-depth from the top surface. As a result, the control perovskite film displayed a smaller variation of E_F from −4.86 eV measured at the top surface to −4.89 eV measured at the buried interface (Figs. S10a and S11a online). In contrast, the target perovskite film exhibited larger variation of E_F between the top and bottom surface (Figs. S10b and S11b online) and gradually downshifted E_F values from the bulk interior (−4.75 eV measured at a depth of ∼250 nm, Fig. S11d online) to the buried interface (−5.01 eV, Fig. S10b online), indicating more p-type characteristics of the film bottom. This could be attributed to the preferable distribution of 2PACz on the bottom surface of perovskite film, which is in accordance with the TOF-SIMS result (Fig. S9 online). Specifically, how the 2PACz modification changed the E_F can be understood and explained by the following two aspects: (1) 2PACz-assisted targeted interaction and passivation could retard the fast oxidation of Sn^{2+} into Sn^{4+}, thus reducing the numbers of Sn vacancies and altering the electronic properties of resultant Sn-Pb perovskite film; (2) appropriate alloying of p-type 2PACz and Sn-Pb perovskite is able to reconfigure the position of E_F [47]. In our case, these two aspects synergistically affected the optoelectronic properties (i.e., energy levels and doping levels, etc.) and eventually, downshifted the E_F. To validate this, we studied the

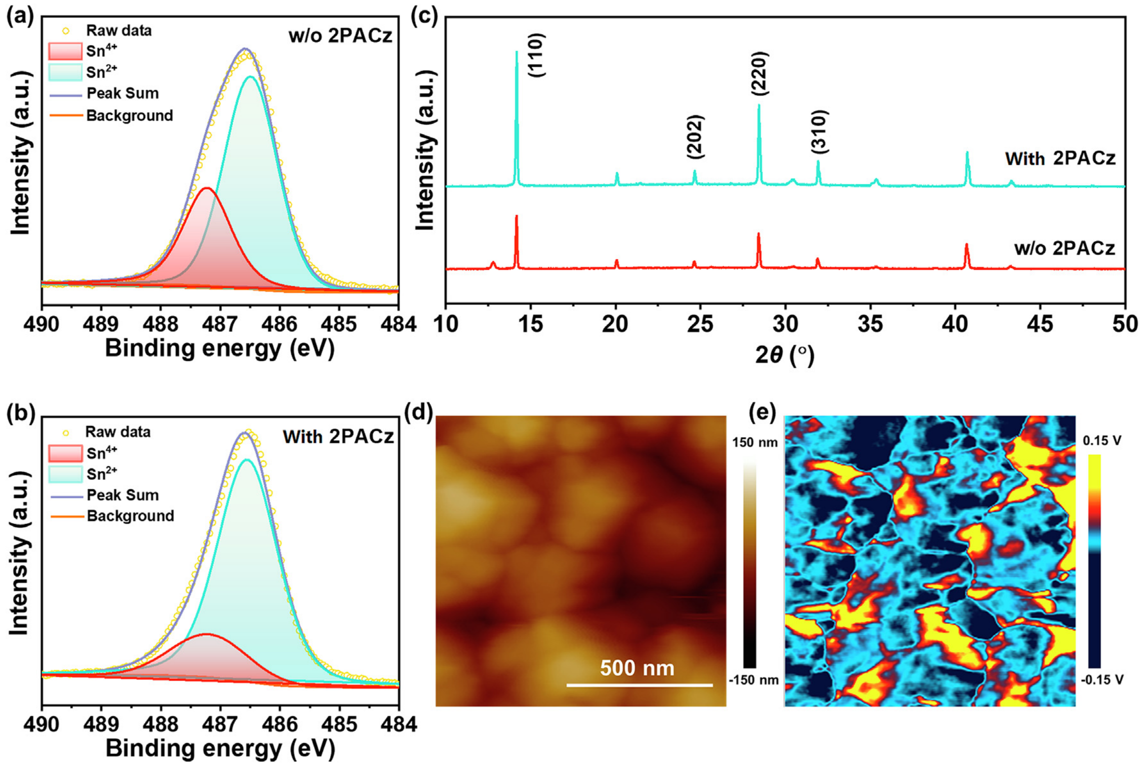

Fig. 2. (Color online) The antioxidation function and spatial distribution of 2PACz in perovskite films. The Sn $3d_{5/2}$ XPS spectra of (a) the control and (b) the 2PACz-modified perovskite films. (c) XRD patterns of the perovskite films without or with 2PACz modification. (d) AFM topographic image and (e) AFM-IR absorption image of 2PACz-modified perovskite films.

effect of different 2PACz concentrations on modulating the Femi levels of resultant perovskite films. Increasing the 2PACz concentrations from 0.1 mol% to 0.2 mol% further down-shifted the E_F to −5.32 eV (Fig. S12 online). Even though the 0.2 mol% 2PACz-doped perovskites become more p-type, the resultant PSC did not observe further PCE enhancement (Fig. S13, Table S1 online), which can be possibly attributed to over-aggregation of 2PACz molecules that could induce additional charge transfer resistance. Hence, we concluded the concentrations and molecular packing densities of 2PACz should be carefully considered and balanced to obtain further performance enhancement. Besides, the valence band maximum (VBM) of the control and 2PACz-modified perovskite films were determined to be −5.50 and −5.46 eV, respectively [48]. Considering the ~1.22 eV bandgap of our employed mixed Sn-Pb perovskite [29], the energy levels of both two perovskite films have been schematically illustrated in Fig. 3a. Obviously, one can expect facilitated interfacial charge extraction from 2PACz-modified perovskite layer to adjacent electrode/contact owing to the well-optimized energy level alignment at relevant interface [49,50]. We also employed KPFM to characterize the surface potentials of the exfoliated mixed Sn-Pb perovskite films with or without 2PACz modification (Fig. 3b). Typically, the contact potential difference (CPD) which is defined as $(\Phi_{tip}-\Phi_{sample})/e$, could reflect the work function of the samples. On one hand, the mean surface potential of the 2PACz-modified perovskite film is higher than that of the control one (Fig. 3c and e), manifesting a higher work function, corresponding to a more p-type (i.e., p^+) characteristic induced by 2PACz modification, which is consistent with the abovementioned UPS result. On the other hand, one can observe the remarkable fluctuation of the CPD for the control perovskite film during linear potential scanning (Fig. 3d). Considering that the KPFM measurement was conducted in ambient air, it is reasonable to speculate that the considerable potential variation of the exfoliated pristine perovskite bottom surface was possibly caused by the fast oxidization of Sn^{2+} into Sn^{4+}, thus strengthening the self p-type doping effect. However, such an uncontrolled p-doping is at the cost of forming numerous Sn vacancies in the perovskite crystal lattice, which would bring about additional charge-trapping sites and induce severe charge recombination [51,52]. As for the 2PACz-modified sample, the CPD has been stabilized between −40 and −60 mV across the film (Fig. 3f), again verifying the exceptional antioxidation ability of the 2PACz additives.

It is acknowledged that the photo-excited charge transfer process at the relevant interfaces of perovskite devices that occurs on a timescale of picoseconds could notably affect interfacial carrier extraction and collection [53]. The fs-TA spectroscopy measurement was performed to understand the ultrafast photo-excited carrier dynamics of the stacked ITO/perovskite/C_{60} semi-devices. Upon fs-laser excitation from the ITO side, both of the control and 2PACz-modified perovskite films displayed distinct ground-state bleaching (GSB) peak around 940 nm in their contour plot of the TA spectra, the intensity of which could reflect the photoinduced carrier densities in the valence band and conduction band of perovskite [54]. In comparison with the control perovskite-based sample, a more promptly faded signal was detected in the contour plot of the stacked film based on 2PACz-modified perovskite (Fig. 4a and b), indicating the photogenerated carriers (i.e., electrons and holes) were more quickly extracted out to both the ITO and C_{60}, respectively. We also investigated the dynamic evolution of GSB signal probed at different delay times (Fig. S14 online), which could reflect the charge carrier dynamics within the stacked devices [55]. Notably, the device based on 2PACz-modified perovskite film demonstrated a much weaker GSB intensity at the same delay time than the control counterpart, indicating faster charge transfer from 2PACz-modified perovskite film to the adjacent contacts. In addition, we noticed the

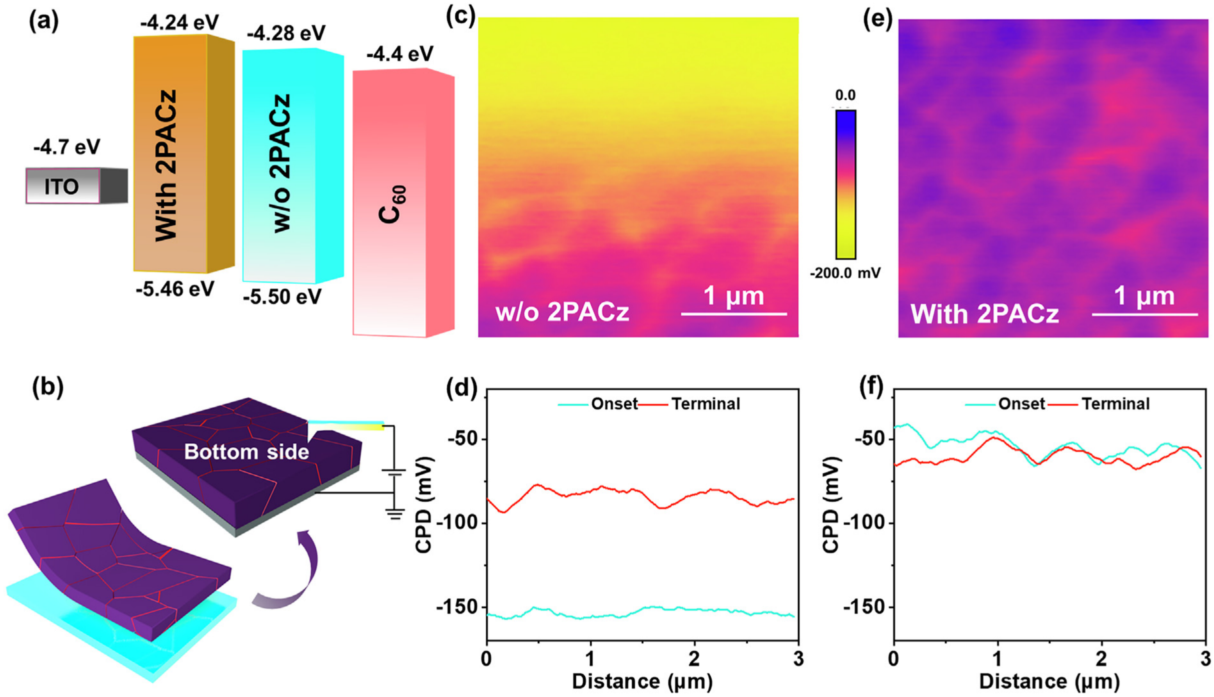

Fig. 3. (Color online) 2PACz modification on energy level and work function of perovskite films. (a) Energy level diagram of the mixed Sn-Pb perovskite films without or with 2PACz modification. (b) The device configuration and setup for KPFM measurement. KPFM image and CPD linear profile of (c, d) the control perovskite film and (e, f) the 2PACz-modified perovskite film. Note, for each sample, two cycles of scan have been conducted to confirm the reliability.

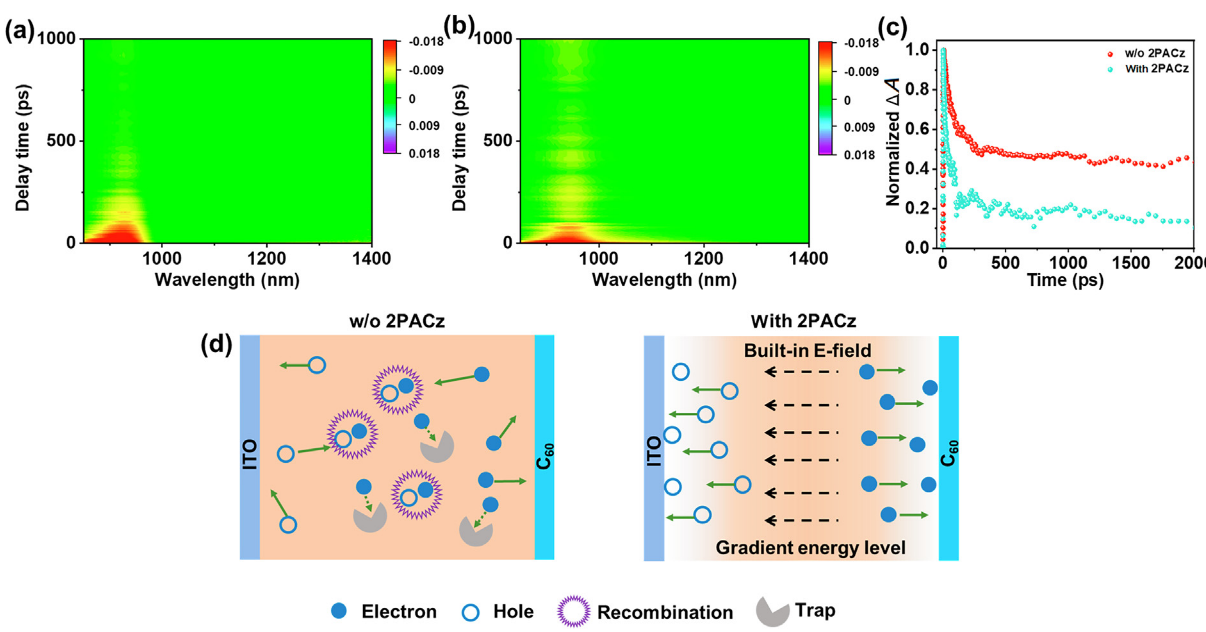

Fig. 4. (Color online) Carrier dynamics of 2PACz-modified mixed Sn-Pb perovskite film and its relevant interfaces. The color TA spectra of semi-devices based on (a) the control perovskite film and (b) the 2PACz-modified perovskite film. (c) TA decay as a function of delay time for the semi-devices based on corresponding perovskite films. (d) Schematic illustration of carrier dynamics in ITO/mixed Sn-Pb perovskites without or with 2PACz modification/C_{60} interface showing the charge separation, extraction and collection.

red-shift of GSB peaks for both control and 2PACz-modified perovskite samples when increasing the delay time. This can be explained by the fact that when the perovskite samples were photoexcited, the density of electrons in the conduction band increased, followed by Coulombic interactions between the electrons of conduction band and increased electron impurity scattering, leading to the renormalization of bandgap. As a result, the perovskite bandgap decreased and the GSB signals red-shifted with time [56,57]. Subsequently, we analyzed and illustrated the decay kinetics of the GSB signals, which demonstrated a faster decay for the ITO/2PACz-modified perovskite/C_{60} sample than that of the ITO/pristine perovskite/C_{60} (i.e., 1096 vs. 1601 ps, Fig. 4c and Table S2 online). This result highlighted the great benefits of targeted 2PACz passivation and energy level modulation on expediting carrier extraction and collection from perovskite layer to the respective electrodes, especially the hole extraction from perovskite layer to ITO substrate, owing to the reinforced built-in electric field for boosted charge separation within perovskite and/or at relevant heterojunction interface, as well as well-aligned interfacial energy level at the ITO/2PACz-modified perovskite interface for facilitated carrier extraction through an ultrafast cascaded shuttle. Such an intriguing and spontaneous interfacial optimization would effectively retard the interfacial carrier recombination, thus ensuring high device performance even in the absence of HTL. The transient photocurrent test of the control and 2PACz-modified devices was performed, in which the target device showcased a faster photocurrent decay than the control device (i.e., 1070 vs. 1170 ns), further verifying faster carrier extraction for the former (Fig. S15 online). The steady-state and TRPL spectra were measured for both the control and 2PACz-modified perovskite films deposited on ITO glass substrates. As compared to the control sample, the 2PACz-modified perovskite film has witnessed a more prominent PL quenching and comparatively fast PL decay lifetime (Fig. S16 online), also suggesting faster photo-induced carrier transfer from the modified perovskite film to the ITO glass substrate, which can be ascribed to the optimized interfacial energy level alignment at the ITO/perovskite interface upon 2PACz modification [58]. Besides, we also measured the PL and TRPL spectra of the control and 2PACz-modified perovskite films deposited on the glass substrates. Compared to the control perovskite film, the 2PACz-modified perovskite film showed enhanced PL intensity and prolonged PL decay lifetime (Fig. S16 online), suggesting the well-passivated defects and the suppressed non-radiative recombination for the target perovskite film.

Though the perovskite materials have ambipolar charge transport characteristics, the photo-generated charge carriers would transport randomly in the perovskite layer if there is no contact junction formed with the adjacent CTL, which could result in serious interfacial electron-hole recombination, thus being detrimental to the photovoltaic performance. Herein, we propose the most plausible model for elucidating the charge separation, extraction and collection within the 2PACz-modified perovskite film and its relevant interface, as is illustrated in Fig. 4d. Specifically, 2PACz molecules have been proved to chemically bond on the GBs of mixed Sn-Pb perovskite crystals, which could spontaneously form an intriguing p/p$^+$ homojunction between intrinsic grain interiors and 2PACz-anchored GBs [59]. This is conducive to dramatically enhancing the internal built-in electric field for boosting the separation of photo-induced carriers, which is especially important in HTL-free devices, as the absence of hole-selective contact would cause unexpected charge accumulation at the ITO/perovskite interface and incur severe charge recombination. In addition, the 2PACz itself is an advantageous hole transport material with a suitable energy level and hole conductivity. In this case, the self-assembly of the 2PACz around the GBs could form an energy-favorable heterojunction with mixed Sn-Pb perovskites, which enabled faster hole extraction and collection from perovskite to 2PACz. With the assistance of the rational combination of the abovementioned perovskite-perovskite homojunction and perovskite-additive heterojunction, favorable cascaded charge transfer can be realized at the integrated interface, which remarkably expedited the hole extraction and collection from perovskite to ITO glass even in the absence of HTL. This effective and efficient spatial separation of holes and electrons beneficially balances the carrier transportation to the respective electrodes and impedes the recombination of free carriers, which is promising to improve the optoelectronic properties and photovoltaic performances of the HTL-free mixed Sn-Pb PSCs. Previous reports demonstrated the utilization of 2PACz or [2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl] phosphonic acid (MeO-2PACz) monolayer as the interfacial layer to modify ITO or fluorine-doped tin oxide (FTO), which just focused on showing the capability of 2PACz or MeO-2PACz to modify the interfacial energy level alignment and act as alternative HTLs (rather than PTAA or NiO_x) to extract the holes from perovskite layer to ITO or FTO electrodes. Specifically, the preparation of 2PACz monolayer requires complicated post-washing treatment to remove excessive 2PACz molecules. In our work, 2PACz was used as an additive in the perovskite ink, we demonstrated that 2PACz could chemically anchor on the GBs and preferably interact with Sn^{2+} ions, which simultaneously facilitated the charge extraction and collection without requiring to deposit an independent HTL, and effectively passivate the defects at GBs and/or film surface, thus reducing both bulk and interfacial non-radiative recombination loss. It is anticipated that the surface of the ITO could not be modified with the similar "all-in-one" method by simply depositing a layer of 2PACz, since it is also required to facilitate the charge separation within the bulk of ~600 nm thick perovskite film, especially in the absence of HTL. In our proposed model, the targeted modification of GBs would provide additional momentum to promote spatial charge separation and transport within the bulk of perovskite film, and facilitate the interfacial hole extraction from perovskite layer to ITO electrode.

To evaluate the impact of targeted passivation and optimized interfacial carrier dynamics on improving the photovoltaic performance of simplified devices, we constructed the HTL-free mixed Sn-Pb PSCs with an inverted p-i-n architecture of ITO/perovskite/ETL/Cu. Specifically, ~600 nm thick perovskite layer was directly deposited on the plasma-treated ITO glass, followed by thermal evaporation of ETL and metal counter electrode (Fig. 5a). As illustrated in Fig. 5b and Table 1, the control PSC without 2PACz modification showcased a low PCE of 15.33%, accompanied by a short circuit current density (J_{sc}) of 30.54 mA/cm^2, an open-circuit voltage (V_{oc}) of 0.75 V and a fill factor (*FF*) of 66.9% measured in the reverse scan, while only achieved a PCE of 14.03% with a J_{sc} of 27.83 mA/cm^2, a V_{oc} of 0.76 V and a *FF* of 66.3% measured in the forward scan, indicating an obvious hysteresis effect (Fig. S17 and Table S3 online). We inspected the impact of 2PACz concentrations (varying from 0.05 mol% to 0.20 mol%) on the device performances of HTL-free mixed Sn-Pb PSCs. The optimal 2PACz concentration has been identified to be 0.10 mol% (Fig. S13 and Table S1 online). Too low dopant concentration was not efficient enough to modulate the energy level of perovskite film to an appropriate extent, while too high dopant concentration would inevitably induce over-aggregation and packing of 2PACz molecules on the surface of perovskite film that would in turn increase the charge transfer resistance and even cause unexpected charge recombination. Encouragingly, a notable performance improvement arising from all photovoltaic parameters (i.e., J_{sc}, V_{oc}, *FF* and PCE) was achieved for the 0.1 mol% 2PACz-modified device (Fig. S18 online). The champion device showed a PCE of 20.21%, accompanied by a J_{sc} of 32.22 mA/cm^2, a V_{oc} of 0.87 V and a *FF* of 72%, with almost negligible hysteresis (Fig. S19 and Table S3 online). To the best of our knowledge, this is the record efficiency for the narrow bandgap mixed Sn-Pb PSCs without HTLs. Compared to the control PSC, the target device displayed enhanced external quantum efficiency (EQE) values over the wavelength regions from 400 to 900 nm, agreeing well with the higher J_{sc} for the latter. The integrated J_{sc} derived from the EQE spectrum is 30.70 mA/cm^2 for the 2PACz-modified device, which is in good accordance with the J_{sc} obtained from the *J*–*V* test, exhibiting a discrepancy of less than 5% (Fig. S20 online) [7,29]. The stabilized power output of 2PACz-modified device was recorded by holding the device at a maximum power point of 0.69 V under continuous AM 1.5 G one-sun illumination, giving a stabilized PCE of 19.5% (Fig. 5c). The PCE histogram of the HTL-free mixed Sn-Pb PSCs without or with 2PACz modification was demonstrated in Fig. 5d,

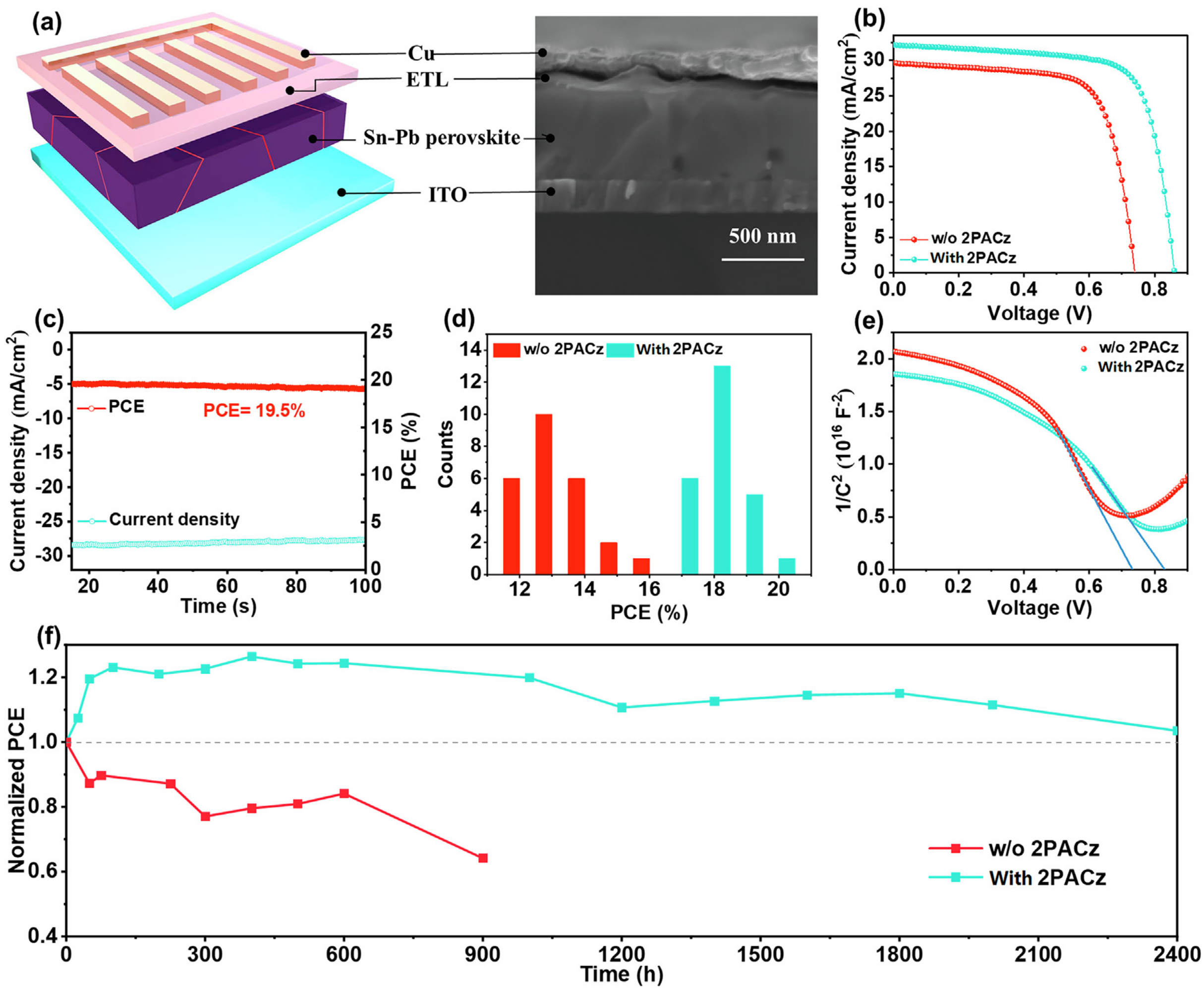

Fig. 5. (Color online) Device photovoltaic performance. (a) Schematic illustration and cross-sectional SEM image of the completed HTL-free mixed Sn-Pb PSC. (b) *J–V* curves of the HTL-free mixed Sn-Pb PSCs without or with 2PACz modification. (c) Steady-state current density and power output of the champion device measured at MPP condition. (d) PCE histogram of the control and 2PACz-modified PSCs. (e) Mott-Schottky plots of the PSCs without or with 2PACz modification. (f) Long-term stability of the control device and 2PACz-modified device, which were stored in an N_2 glovebox without encapsulation and tested in ambient air ((25 ± 5) °C and 35% ± 10% relative humidity).

Table 1
Summary of the photovoltaic parameters of the HTL-free mixed Sn-Pb PSCs without or with 2PACz modification under AM 1.5G one sun illumination.

PSCs	J_{sc} (mA/cm^2)	V_{oc} (V)	PCE (%)	Average PCE (%)[a]	*FF* (%)
w/o 2PACz	30.54	0.75	15.33	12.80 ± 1.01	66.9
With 2PACz	32.22	0.87	20.21	18.39 ± 0.69	72.1

[a] The average PCE was calculated from a batch of 25 cells for each condition.

in which the data was collected from a batch of 25 devices for each condition. In contrast to the control devices, the targeted devices exhibited narrower PCE distribution. More than 76% of the targeted devices with 2PACz modification obtained PCE above 18%, while only about 12% of the control counterparts had PCEs over 14%. This outcome highlighted the effectiveness and significance of 2PACz-assisted targeted modification in enhancing the device performance, accompanied by a remarkably enhanced reproducibility of the PSCs even in the absence of HTL. We also fabricated the mixed Sn-Pb PSCs based on the independent 2PACz HTL fabricated with various concentrations of 2PACz solution, which only showed an optimized PCE of 16.50%, with a J_{sc} of 29.63 mA/cm^2, a V_{oc} of 0.81 V and an *FF* of 68.7% (Fig. S21 online). This performance just slightly outperformed the control HTL-free devices, but it is largely inferior to that of the 2PACz-incorporated HTL-free PSCs. Similarly, the 2PACz analogue, namely, MeO-2PACz, was also utilized to prepare HTL-free PSCs and independent MeO-2PACz HTL-based PSCs. The results also demonstrated better device performance of MeO-2PACz doped HTL-free PSCs than that of the independent MeO-2PACz HTL-based counterparts, though both of these two devices exhibited inferior device performances than the 2PACz-modified HTL-free PSCs (Fig. S22 online). Obviously, for selecting other alternative carbazole phosphonic acid analogues, one should comprehensively consider the quality of resultant thin films (i.e., packing density, uniformity and thickness, etc.) and corresponding optoelectronic properties (i.e., energy level and mobility, etc.), which would ensure high device performances. Accordingly, our work provides a perfect addition to the family of CTL-free PSCs and demonstrates a record PCE (20.21%) for the Sn-based HTL-free PSCs. The PCE of our device is comparable to or even much higher than that of the conventional devices with HTL, which implies average PCEs >19%. In addition, it is generally believed that the efficiencies of Sn-based PSCs are still inferior to that of Pb-based counterparts. We demonstrated that the Sn-based HTL-free PSCs can work as efficiently as the Pb-based HTL-free PSCs, which represents a significant advance over the state-of-the-art devices (Table S4 online), especially considering the narrow bandgap PSCs are promising to be used in perovskite-based tandem solar cells, which could bring a huge economic benefit if the HTL-free concept can be applied.

As discussed above, the improvement of J_{sc} and *FF* can be mainly explained by the well-optimized interfacial energy level alignment (Fig. 3a), and strengthened built-in potential (from 730 to 830 mV, Fig. 5e), thus reducing the charge transport resistance and facilitating the carrier extraction [20]. One can also notice the remarkable enhancement of V_{oc} by 120 mV for the HTL-free device simply with 2PACz passivation, which can be ascribed to the decreased defect density and largely suppressed charge recombination. To gain deep insight into the defect profile of the PSCs, SCLC measurement was applied. As illustrated in Fig. S23 (online), the dark *J*–*V* curves of the hole-only devices consisted of three regions, namely, the linear Ohmic region at low bias, the trap-filling region and the space charge-limited region. The trap densities were determined by the trap-filled limit voltage (V_{TFL}) governed by the Mott-Gurney law:

$$V_{TFL} = \frac{en_tL^2}{2\varepsilon\varepsilon_0,}$$

where ε_0 is the vacuum permittivity, ε is the relative dielectric constant, L is the perovskite film thickness, n_t is the trap density and e is the elementary electronic charge [60–62]. A smaller V_{TFL} and trap densities were observed for the device comprising 2PACz, as compared to the control counterpart, manifesting that the incorporation of 2PACz greatly reduced the defect density and minimized the trap-assisted recombination loss owing to the targeted passivation, as well as enhanced stabilization and antioxidation of Sn^{2+} ions. Electrochemical impedance spectroscopy (EIS) measurement was carried out for PSCs fabricated without or with 2PACz to further investigate the charge recombination behaviors. The Nyquist inset plots were fitted using the equivalent circuit model depicted in the inset of Fig. S24 (online). As a result, the recombination resistance (R_{rec}) for the 2PACz-modified device is larger than that of the control device without any modification. This result reflected the advantageous role of 2PACz passivation in suppressing the non-radiative recombination within the devices, which can be attributed to the well-optimized optoelectronic properties and well-modulated interfacial carrier dynamics of 2PACz-modified devices, namely, facilitated interfacial hole extraction at the ITO/perovskite interface owing to the facilitated charge separation and reduced charge transfer resistance induced by 2PACz modification.

Aside from efficiency, stability is another critical aspect to evaluate the performance of photovoltaic devices, especially for the Sn-based PSCs. In this work, long-term stability of the HTL-free mixed Sn-Pb PSCs was traced and investigated. The PSCs without encapsulation were stored in an inert gas environment at room temperature. The performance test was conducted in ambient air with 35% ± 10% relative humidity. As demonstrated in Fig. 5f, the PCE of the control device degraded to 64% of its initial PCE value after 900 h, whereas the device with 2PACz stabilization demonstrated strikingly enhanced stability, and exhibited almost no PCE loss even after 2400 h, which is comparable to or even outperforms than other state-of-the-art mixed Sn-Pb PSCs (Table S5 online). During the aging process, the targeted passivation of Sn-Pb perovskites by 2PACz played a more important role in extending the lifespan, which can be attributed to the improved quality of perovskite films with enhanced crystallinity, reduced defects and reinforced antioxidation capability. Moreover, Fig. 5f highlights a notable difference in the PCE evolution of the control and target PSCs during the first 100 h. For the control device without 2PACz modification, the PCE degradation can be primarily attributed to the easy oxidation and potential decomposition of mixed Sn-Pb perovskites, particularly the unencapsulated devices which were subjected to ambient conditions during *J*–*V* test, thus aggregating the decline in efficiency. Conversely, the target device exhibited an upward trend in PCE during the same time frame due to the effective passivation and anti-oxidation capabilities of 2PACz. This finding aligns with previous studies, which have demonstrated that the well-passivated PSCs require appropriate aging, such as grain coarsening and/or self-healing of perovskite films, to maximize their photovoltaic performances [5,16]. In addition to the shelf-life stability, we also tested the thermal stability of devices and performed the continuous MPP tracking (at 0.67 V) of devices under one-sun illumination. Specifically, the 2PACz-modified device demonstrated enhanced thermal stability, which maintained ~85% of its initial PCE after heating at 60 °C for 200 h. In contrast, the control device quickly degraded to 64% of its original PCE upon heating for 100 h (Fig. S25 online). Moreover, the target device retained ~86% of its initial PCE after continuous one-sun illumination for 150 h, while the control counterpart remained only ~50% of the original PCE under the same light-soaking conditions (Fig. S26 online). Overall, the improved long-term stability of the modified device could be ascribed to multifaceted roles of 2PACz modification, namely, improving the antioxidation capability of Sn-based perovskite film (Fig. 2a, b), enhancing defect passivation efficacy (Fig. S23 online) and optimizing interfacial carrier dynamics (Fig. 4a–c, Fig. S24 online).

4. Conclusion

In summary, we have demonstrated a simple chemical modification way to prominently enhance the optoelectronic properties and photovoltaic performances of the mixed Sn-Pb PSCs with simplified device structures via carefully controlling the coordination chemistry and modulating the interaction manner. The multifunctional 2PACz molecule can strongly coordinate with Sn^{2+} ions in a preferable way, retard them from being oxidized into Sn^{4+}, as well as passivate the crystallographic defects and reduce carrier recombination. In addition, 2PACz modification is beneficial to optimize the contact junctions, reinforce the built-in potential, and induce favorable energy level cascade, thus accelerating the spatial separation, directional transport and comprehensive collection of charge carriers from perovskite layer to adjacent contacts. As a result, the modified HTL-free devices attained the best PCE up to 20.21%, accompanied by a high V_{oc} of 0.87 V, featuring outstanding long-term stability. This is the highest efficiency for HTL-free mixed Sn-Pb PSCs reported to date. This work provides a comprehensive understanding of coordination inclination and chemical interaction between exotic molecules and perovskites, which could pave the way for the development of more environmental-friendly, stable PSCs with optimal bandgap, simplified device architecture and high performance in a cost-effective manner.

Conflict of interest

The authors declare that they have no conflict of interest.

Acknowledgments

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Author contributions

Wu-Qiang Wu conceived the idea and supervised the whole project. Xueqing Chang and Jun-Xing Zhong completed the experimental studies including sample preparation, device fabrication, and related characterization. Guo Yang, Ying Tan, Jie Zhou, and Zhibin Yang performed the device measurements. Li Gong performed the KPFM measurements and analysis. Xing Ni, Yujin Ji, and Youyong Li conducted the theoretical calculation. Guodong Zhang, Yifan Zheng, and Yuchuan Shao performed the tDOS measurement. Lianzhou Wang assisted in data analysis. Wu-Qiang Wu took the responsibility of writing, review and editing. All authors discussed the results and commented on the manuscript.

Appendix A. Supplementary materials

Supplementary materials to this article can be found online at https://doi.org/10.1016/j.scib.2023.05.012.

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